Compactly supported Barron functions exist

The two questions in arXiv:2006.05982 are answered by arXiv:2209.01173

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Status. `literature_already_answered`. Later work by Wojtowytsch constructs nonzero, nonnegative, compactly supported Barron functions in every dimension. Consequently both alternatives asked about in the source have an affirmative answer.

Source question

At the end of Section 5.2 of E–Wojtowytsch, *Representation formulas and pointwise properties for Barron functions*, arXiv:2006.05982v2, the authors ask whether there are compactly supported Barron functions, or nonnegative Barron functions decaying faster than $|x|^{-1}$ at infinity [1]. Their Barron norm controls both the spatial weights and the biases; equivalently, it is the total variation of a signed measure on the augmented unit sphere.

Decisive later theorem

Theorem 3.1 of Wojtowytsch, *Optimal bump functions for shallow ReLU networks*, arXiv:2209.01173v1, states that for every odd $D \in \mathbb{N}$ there is a radial function $F_D \in \mathcal{B}(\mathbb{R}^D)$ with

$$F_D(0) = 1, \qquad F_D = 0 \text{ on } \mathbb{R}^D \setminus B_1(0), \qquad 0 \le F_D \le 1.$$

The paper explicitly describes these functions as compactly supported and nonnegative [2]. It also gives a finite representer formula of the form

$$F_D(x) = 1 + \sum_{j=0}^{(D+1)/2} c_j \int_{S^{D-1}} \sigma(\nu^T x - b_j)\, d\mathcal{H}^{D-1}(\nu), \qquad 0 = b_0 < \cdots < b_{(D+1)/2} = 1.$$

Thus the biases are bounded, and the displayed representation has finite *full* Barron norm, not merely finite translation-invariant seminorm.

Norm and dimension checks

The full norm in arXiv:2209.01173 uses the parameter cost $|a|\big(|w|_2 + |b|\big)$, whereas arXiv:2006.05982 uses $|a|\,|(w,b)|_2$. These costs are equivalent because

$$|(w,b)|_2 \le |w|_2 + |b| \le \sqrt{2}\,|(w,b)|_2.$$

Positive homogeneity of ReLU therefore normalizes every nonzero augmented parameter (w, b) to the unit sphere and produces exactly the signed-sphere representation used in the source.

For an arbitrary dimension d, choose $D = d$ when d is odd and $D = d + 1$ when d is even, and put

$$f_d(x) = F_D(x, 0, \dots, 0), \qquad x \in \mathbb{R}^d.$$

Restriction preserves the Barron class: in every ridge term it deletes some spatial coordinates and can only decrease the augmented parameter norm. This restriction property is also stated explicitly immediately after Theorem 3.1 in arXiv:2209.01173. Moreover $f_d(0) = 1$, $0 \le f_d \le 1$, and $f_d(x) = 0$ for $|x| \ge 1$. Hence f_d is a nonzero compactly supported Barron function in every $d \ge 1$.

Because f_d is nonnegative and eventually identically zero, it also decays faster than $|x|^{-1}$ (indeed, faster than every power). The single family therefore answers both questions in arXiv:2006.05982 affirmatively.

Chronology and provenance

The source version is from 2021; arXiv:2209.01173 appeared in September 2022 and was published in JMLR in 2024 (where the result is Theorem 7). The later paper is by the second author of the source. This packet records that later literature answer and makes no novelty claim.

References

[1] W. E and S. Wojtowytsch, *Representation formulas and pointwise properties for Barron functions*, Calc. Var. Partial Differential Equations 61 (2022), Art. 46; arXiv:2006.05982.

[2] S. Wojtowytsch, *Optimal bump functions for shallow ReLU networks: weight decay, depth separation and the curse of dimensionality*, J. Mach. Learn. Res. 25 (2024), 1–49; arXiv:2209.01173.